



Goniometrix Design Roadmap

A user-driven roadmap connecting research insights to concept development and prototyping across four course phases.

Phase 1



Identify

- Problem stance
- 1:1 interviews
- User observation
- Competitive Analysis
- Opportunity Framing



Defined the need for independent, repeatable joint-angle measurement

Phase 2



Understand

- Key user groups
- Pain points + task breakdown
- Core insights
- Design requirements
- System context



Defined the need for independent, repeatable joint-angle measurement

Phase 3



Conceptualize

- 30 initial concepts
- AI-supported concept expansion
- semantic clustering
- Feasibility/novelty/impact mapping
- Dot voting + shortlist



Defined the need for independent, repeatable joint-angle measurement

Phase 4



Realize

- System diagram + feedback loops
- Sensing → prediction → actuation
- Hardware integration
- Desktop GUI prototyping
- Final wearable prototype



Defined the need for independent, repeatable joint-angle measurement



Key Quotes

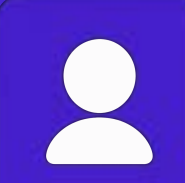
"I can't measure accurately on my own."



"I need feedback I can trust."



"I want to track progress between sessions."



Core Needs

- Independent self-measurement
- Clear, trustworthy feedback
- Repeatable tracking over time
- Low-effort, wearable setup



Design Principles

- Support independent use
- Reduce physical + cognitive load
- Make feedback actionable
- Build trust through consistency and progress tracking



Roadmap Outcome

User research and design principles guided the project from problem framing to a wearable smart goniometer with dual IMU sensing, haptic/audio feedback, and a live companion GUI.



Future direction: validate with more users, refine calibration, and expand longitudinal process tracking